

МОСКОВСКИЙ ГОСУДАРСТВЕННЫЙ ТЕХНИЧЕСКИЙ УНИВЕРСИТЕТ
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Дисциплина “Парадигмы и конструкции языков программирования”

Отчет по лабораторной работе №1

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Листинг программы в процедурной парадигме на языке python:

```
import math
import sys

REQUIRED_NUMBER_OF_ARGUMENTS = 4
MIN_NUMBER_OF_ARGUMENTS = 1

def IsQuantityArgsCorrect():
    if len(sys.argv) > REQUIRED_NUMBER_OF_ARGUMENTS:
        print("Too many arguments entered, please try again" '\n')
        calculate_roots(get_coeff_a(), get_coeff_b(), get_coeff_c())

    elif len(sys.argv) < REQUIRED_NUMBER_OF_ARGUMENTS and len(sys.argv) !=
MIN_NUMBER_OF_ARGUMENTS:
        print("Too few arguments entered, please try again" '\n')
        calculate_roots(get_coeff_a(), get_coeff_b(), get_coeff_c())

    elif len(sys.argv) == MIN_NUMBER_OF_ARGUMENTS:
        calculate_roots(get_coeff_a(), get_coeff_b(), get_coeff_c())

    else:

        coeff_a = Check_correctness_coefficient_from_arg(sys.argv[1], "a")
        coeff_b = Check_correctness_coefficient_from_arg(sys.argv[2], "b")
        coeff_c = Check_correctness_coefficient_from_arg(sys.argv[3], "c")

        calculate_roots(coeff_a, coeff_b, coeff_c)

def get_coeff_a():
    try:
        coeff_a = float(input("Enter coefficient A: "))
        return coeff_a
    except ValueError:
        print(f"Incorrect coefficient A entered. Please try again" + '\n')
        while True:
            try:
                coeff_a = float(input(f"Enter coefficient A: "))
                return coeff_a
            except ValueError:
                print(f"Incorrect coefficient A entered. Please try again" + '\n')

def get_coeff_b():
    try:
        coeff_b = float(input("Enter coefficient B: "))
        return coeff_b
    except ValueError:
        print(f"Incorrect coefficient B entered. Please try again" + '\n')
        while True:
            try:
                coeff_b = float(input(f"Enter coefficient B: "))
                return coeff_b
            except ValueError:
```

```

        print(f"Incorrect coefficient B entered. Please try again" + '\n')

def get_coeff_c():
    try:
        coeff_c = float(input("Enter coefficient C: "))
        return coeff_c
    except ValueError:
        print(f"Incorrect coefficient C entered. Please try again" + '\n')
        while True:
            try:
                coeff_c = float(input(f"Enter coefficient C: "))
                return coeff_c
            except ValueError:
                print(f"Incorrect coefficient C entered. Please try again" + '\n')

def Check_correctness_coefficient_from_arg(coeff, coeff_name):
    try:
        coeff = float(coeff)
        return coeff
    except ValueError:
        print(f"Incorrect coefficient {coeff_name} entered. Please try again" + '\n')
        while True:
            try:
                coeff = float(input(f"Enter coefficient {coeff_name}: "))
                return coeff
            except ValueError:
                print(f"Incorrect coefficient {coeff_name} entered. Please try again"
+ '\n')

def calculate_roots(coeff_a, coeff_b, coeff_c):
    print('\n' + "entered coefficients:" + '\n')
    print(f"coefficient A: {coeff_a}")
    print(f"coefficient B: {coeff_b}")
    print(f"coefficient C: {coeff_c}" + '\n')

    roots = []

    discriminant = coeff_b**2 - 4*coeff_a*coeff_c
    if discriminant < 0:
        print('No roots found')
        return

    else:
        root1 = (-coeff_b + math.sqrt(discriminant)) / (2 * coeff_a)
        roots.append(root1)
        root2 = (-coeff_b - math.sqrt(discriminant)) / (2 * coeff_a)

        if root2 != root1:
            roots.append(root2)
        print(f"found roots --- {roots}")

IsQuantityArgsCorrect()

```

Листинг программы (ООП) на языке python:

```
import math
import sys

REQUIRED_NUMBER_OF_ARGUMENTS = 4
MIN_NUMBER_OF_ARGUMENTS = 1

class Quadratic_Equation:

    def __init__(self, coeff_a: float = 0, coeff_b: float = 0, coeff_c: float = 0):
        self.__coeff_a = coeff_a
        self.__coeff_b = coeff_b
        self.__coeff_c = coeff_c

    def Calculate_roots(self):
        roots = []

        discriminant = self.__coeff_b**2 - 4*self.__coeff_a*self.__coeff_c
        if discriminant < 0:
            print('No roots found')
            exit()

        else:
            root1 = (-self.__coeff_b + math.sqrt(discriminant)) / (2 * self.__coeff_a)
            roots.append(root1)
            root2 = (-self.__coeff_b - math.sqrt(discriminant)) / (2 * self.__coeff_a)

            if root2 != root1:
                roots.append(root2)
            print(f"found roots --- {roots}")
        pass

def From_User_Input_A():
    try:
        coeff_a = float(input("Enter coefficient A: "))
        return coeff_a
    except ValueError:
        print("Incorrect coefficient A entered. Please try again" + '\n')
        while True:
            try:
                coeff_a = float(input("Enter coefficient A: "))
                return coeff_a
            except ValueError:
                print("Incorrect coefficient A entered. Please try again" + '\n')

def From_User_Input_B():
    try:
        coeff_b = float(input("Enter coefficient B: "))
        return coeff_b
    except ValueError:
        print("Incorrect coefficient B entered. Please try again" + '\n')
```

```

        while True:
            try:
                coeff_b = float(input("Enter coefficient B: "))
                return coeff_b
            except ValueError:
                print("Incorrect coefficient B entered. Please try again" + '\n')

def From_User_Input_C():
    try:
        coeff_c = float(input("Enter coefficient C: "))
        return coeff_c
    except ValueError:
        print("Incorrect coefficient C entered. Please try again" + '\n')
        while True:
            try:
                coeff_c = float(input("Enter coefficient C: "))
                return coeff_c
            except ValueError:
                print("Incorrect coefficient C entered. Please try again" + '\n')

def Check_correctness_coefficient_from_arg(coeff, coeff_name):
    try:
        coeff = float(coeff)
        return coeff
    except ValueError:
        print("Incorrect coefficient {coeff_name} entered. Please try again" + '\n')
        while True:
            try:
                coeff = float(input(f"Enter coefficient {coeff_name}: "))
                return coeff
            except ValueError:
                print(f"Incorrect coefficient {coeff_name} entered. Please try again" +
'\n')

def IsQuantityArgsCorrect():
    if len(sys.argv) > REQUIRED_NUMBER_OF_ARGUMENTS:
        print("Too many arguments entered, please try again '\n'")
        equation = Quadratic_Equation(From_User_Input_A(), From_User_Input_B(),
From_User_Input_C())
        equation.Calculate_roots()

    elif len(sys.argv) < REQUIRED_NUMBER_OF_ARGUMENTS:
        print("Too few arguments entered, please try again '\n'")
        equation = Quadratic_Equation(From_User_Input_A(), From_User_Input_B(),
From_User_Input_C())
        equation.Calculate_roots()

    else:
        coeff_a = Check_correctness_coefficient_from_arg(sys.argv[1], "a")
        coeff_b = Check_correctness_coefficient_from_arg(sys.argv[2], "b")
        coeff_c = Check_correctness_coefficient_from_arg(sys.argv[3], "c")

```

```
equation = Quadratic_Equation(coeff_a, coeff_b, coeff_c)
equation.Calculate_roots()

IsQuantityArgsCorrect()
```

Листинг программы в процедурной парадигме на языке go:

```
package main

import (
    "fmt"
    "math"
    "os"
    "strconv"
)

const RequiredNumberOfArguments = 4
const MinNumberOfArguments = 1
const FloatBitSize = 64

func GetCoeffA() float64 {
    for {
        fmt.Print("Enter coefficient a: ")
        var coeffA string
        fmt.Scan(&coeffA)

        a, err := strconv.ParseFloat(coeffA, FloatBitSize)
        if err != nil {
            fmt.Println("Incorrect coefficient A entered. Please try again")
        } else {
            return a
        }
    }
}

func GetCoeffB() float64 {
    for {
        fmt.Print("Enter coefficient b: ")
        var coeffB string
        fmt.Scan(&coeffB)

        b, err := strconv.ParseFloat(coeffB, FloatBitSize)
        if err != nil {
            fmt.Println("Incorrect coefficient B. Please try again")
        } else {
            return b
        }
    }
}

func GetCoeffC() float64 {
    for {
```

```

        fmt.Print("Enter coefficient c: ")
        var coeffC string
        fmt.Scan(&coeffC)

        c, err := strconv.ParseFloat(coeffC, FloatBitSize)
        if err != nil {
            fmt.Println("Incorrect coefficient C. Please try again")
        } else {
            return c
        }
    }
}

func CheckCorrectnessCoeffFromArg(rawCoeff, coeffName string) float64 {

    coeff, err := strconv.ParseFloat(rawCoeff, FloatBitSize)
    if err == nil {
        return coeff
    } else {
        for {
            fmt.Printf("Incorrect coefficient %s. Please try again", coeffName)
            fmt.Println()

            fmt.Printf("Enter coefficient %s: ", coeffName)
            var coeff string
            fmt.Scan(&coeff)

            finalCoeff, err := strconv.ParseFloat(coeff, FloatBitSize)
            if err == nil {
                return finalCoeff
            }
        }
    }
}

func ShowAllCoeff(a, b, c float64) {
    fmt.Println("Entered coefficients:")
    fmt.Println()
    fmt.Printf("a = %f", a)
    fmt.Println()
    fmt.Printf("b = %f", b)
    fmt.Println()
    fmt.Printf("c = %f", c)
    fmt.Println()
    fmt.Println()
}

func CalculateRoots(a, b, c float64) {
    ShowAllCoeff(a, b, c)

    discriminant := math.Pow(b, 2) - 4*a*c

    var roots []float64

```

```

    if discriminant < 0 {
        fmt.Println("No roots found")
        os.Exit(0)
    } else {
        root1 := (-b + math.Sqrt(discriminant)) / (2 * a)
        roots = append(roots, root1)
        root2 := (-b - math.Sqrt(discriminant)) / (2 * a)

        if root1 != root2 {
            roots = append(roots, root2)
        }
        fmt.Println("found roots ---", roots)
    }
}

func CalculateWithUserEnter() {
    a := GetCoeffA()
    b := GetCoeffB()
    c := GetCoeffC()

    CalculateRoots(a, b, c)
}

func main() {
    if len(os.Args) > 4 {
        fmt.Println("Too many arguments entered")
        fmt.Println()
        CalculateWithUserEnter()

    } else if len(os.Args) < 4 {
        fmt.Println("Too few arguments entered")
        fmt.Println()
        CalculateWithUserEnter()

    } else {
        a := CheckCorrectnessCoeffFromArg(os.Args[1], "A")
        b := CheckCorrectnessCoeffFromArg(os.Args[2], "B")
        c := CheckCorrectnessCoeffFromArg(os.Args[3], "C")

        CalculateRoots(a, b, c)
    }
}

```

Результат выполнения программ


```
koroligor@MacBook-Pro-Igor Labs3sem % python3 Lab1.py
Enter coefficient A: 2
Enter coefficient B: -1
Enter coefficient C: 2

entered coefficients:

coefficient A: 2.0
coefficient B: -1.0
coefficient C: 2.0

No roots found
```